

Aditya Srinivasa Rao Udutha

Data Engineer

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SUMMARY

- Experienced Data Engineer with over four years of expertise in designing, implementing, and optimizing data pipelines and big data solutions.
- Proficient in Python, SQL, and ETL tools like Informatica and SSIS, ensuring data consistency, reliability, and efficient data processing across various platforms.
- Expert in Big Data technologies including Hadoop, MapReduce, and Databricks, enabling the processing and analysis of large datasets to generate actionable insights.
- Skilled in Cloud Technologies such as Azure Data Lake, Cosmos DB, and AWS services (S3, Lambda) for data storage, management, and scalable processing workflows.
- Advanced Analytics and Machine Learning experience with tools like Scikit-learn, Seaborn, and TensorFlow, enhancing predictive analytics and supporting strategic decision-making.
- Developed Interactive Dashboards using Power BI and Tableau, providing real-time data visualization and reporting capabilities to stakeholders.
- Strong background in statistical analysis and data visualization using R and Matplotlib, uncovering trends and insights to support data-driven decisions.
- Collaborative and Agile-focused, working with cross-functional teams to integrate data from various systems, ensuring data consistency and seamless workflows.

PROFESSIONAL EXPERIENCE

Social Work (Remote) USA

Data Engineer

Feb 2024 - Current

- Led the development of scalable ETL pipelines using Python (Pandas, SQLAlchemy) and Informatica to ingest, transform, and load customer data from various sources into PostgreSQL databases, ensuring high data consistency and reliability, which improved customer insights and data-driven marketing strategies.
- Designed and implemented a big data solution using Hadoop HDFS and MapReduce to process and analyze streaming data from IoT devices, enabling real-time analytics and insights that enhanced product performance monitoring and predictive maintenance capabilities.
- Utilized Azure Data Lake and Cosmos DB to store and manage financial data, optimizing data retrieval processes and ensuring high availability and scalability, which supported timely and accurate financial reporting and compliance with industry regulations.
- Leveraged Databricks and Scikit-learn to develop machine learning models that forecasted sales trends, utilizing Seaborn for data visualization to present findings in Power BI dashboards, which provided actionable insights to stakeholders and informed strategic sales decisions.

Dell Technologies India

Data Engineer

Jan 2020 – July 2022

- Developed and maintained scalable ETL pipelines using Python (Pandas, SQLAlchemy) and Informatica, to efficiently ingest, transform, and load data from multiple sources into PostgreSQL databases, ensuring data consistency and reliability across the organization.
- Implemented big data solutions using Hadoop HDFS and MapReduce to process and analyze large datasets, enabling efficient data processing and generation of actionable insights for technological advancements.
- Utilized Azure Data Lake and Cosmos DB for data storage and management, optimizing data retrieval processes and ensuring high availability and scalability for enterprise-level applications, which supported critical business operations.
- Leveraged Databricks for advanced data processing and machine learning tasks, utilizing Scikit-learn for model development and Seaborn for data visualization, thereby enhancing predictive analytics capabilities and supporting strategic decision-making.
- Designed and developed interactive dashboards using Power BI, incorporating DAX for advanced calculations and Power Query for data transformation, providing real-time data visualization and reporting capabilities to stakeholders.
- Conducted statistical analysis and data visualization using R and Matplotlib, to uncover trends and insights in technological data, supporting innovation initiatives and enabling data-driven decision-making across various departments.
- Collaborated with cross-functional teams in an Agile environment to integrate data from various systems and platforms, ensuring data consistency and accuracy across the organization, and facilitating seamless data workflows.
- Optimized database performance by implementing indexing, partitioning, and query optimization techniques in PostgreSQL, which enhanced data retrieval speeds and overall system efficiency, improving user experience and operational productivity.
- Implemented robust data security and compliance measures by leveraging Azure security features, ensuring the protection of sensitive data and adherence to regulatory requirements, thereby maintaining data integrity and confidentiality.

- Automated data workflows and monitoring using Python scripts, improving operational efficiency and significantly reducing manual intervention in data processing tasks, which allowed for more focus on strategic analysis and innovation.

Cipla

Data Analyst

Feb 2019 – Dec 2019

- Utilized Python, NumPy, and Pandas to analyze and process large datasets of patient and insurance claims data, ensuring data accuracy and regulatory compliance, which enhanced decision-making in healthcare management.
- Developed and maintained ETL processes using SQL to extract, transform, and load healthcare data from various sources into MySQL databases, ensuring efficient and reliable data integration and management.
- Designed and implemented interactive Tableau dashboards, providing real-time visualization of patient outcomes and insurance claims, enabling stakeholders to monitor key performance indicators and make data-driven decisions.
- Leveraged AWS services such as S3 and Lambda to store and process healthcare data, optimizing data storage solutions and enabling scalable data processing workflows.
- Conducted statistical analysis and data visualization using Matplotlib to uncover trends and insights in healthcare data, supporting strategic initiatives and improving patient care outcomes.
- Collaborated with cross-functional teams using Git for version control to manage and track changes in data analysis scripts, ensuring seamless integration and collaboration across healthcare projects.

TECHNICAL SKILLS

Programming Language:	R, Python, SQL, Scala
Packages:	NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, Seaborn, TensorFlow
Reporting Tools:	Tableau, Power BI, SSRS, Google Data Studio
ETL Tools:	SSIS, Informatica, Matillion, Fivetran, Coalesce, DBT
Big Data Ecosystem:	Hadoop, MapReduce, Hive, Apache Spark, Pig, Sqoop, Kafka, Kinesis, Airflow
Database:	PostgreSQL, MySQL, PL/SQL, NoSQL - MongoDB, T-SQL
Cloud Technologies:	Azure (Data Lake, Data Factory, Databricks, SQL Database, Synapse Analytics, Blob Storage, Functions, Cosmos DB, HDInsight), AWS (S3, EC2, Redshift, Lambda, Kinesis, QuickSight, IAM Policies, Athena, DynamoDB, Elasticsearch)
Other Tools:	Git, MS Office, Docker, Kubernetes, Maven, Jenkins
Methodology	SDLC, Agile, Waterfall
IDE’s:	PyCharm, Jupyter Notebook
Operating Systems:	Windows, Linux

EDUCATION

Texas A&M University Commerce, Texas

Master of Science in Computer Science

Aug 2022 - May 2024

Mallareddy Engineering College Hyderabad, India

Bachelor’s in Electronics & Communication

Aug 2016 – Jun 2020

PROJECTS

Color and Texture Based Approach for Web Image Retrieval Using Deep Learning Techniques:

- Retrieving the similar image from the data base using MATLAB Software/Image Processing by using the Unified color and intensity level matrices and grey level matrices methods.

Vehicle Number Plate Recognition Software Using Machine Learning:

- Retrieving the number plate from the input image using Machine Learning algorithms.